

REMNANT FIELDWORKS — WITNESS SERIES — FIRST RECORD

WITNESS-1 Results Report

Quantum-Sourced Authorization Nonces with Verifiable Provenance

Series	WITNESS (first record)
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Execution Date	2026-07-18 (UTC)
DOI (version)	https://doi.org/10.5281/zenodo.21424324
DOI (concept)	https://doi.org/10.5281/zenodo.21424323

✓ **OVERALL VERDICT: PASS**

Mandatory Scope Disclaimer

“Results apply within the tested circuit model, backend, qubits, calibration, shot counts, software harnesses, and stated experimental conditions. This is an experimental evidence record, not a universal security proof or production certification.”

Harness Fix Disclosure (pre-execution, mandatory per RF protocol)

Two corrections were made to the harness before execution. Neither touches the circuit, procedure, criteria, thresholds, or oracle:

1. **Channel name:** `ibm_quantum` → `ibm_quantum_platform` (API string changed in qiskit-ibm-runtime 0.48.0; old string rejected at connection time).
2. **Execution mode:** `Session` → `Batch` (Session not available on the open plan; Batch produces identical circuit execution semantics for single-job submissions).

These fixes are disclosed here and in `results/raw/job_meta.json`. The `MANIFEST.sha256` covers the pre-fix code; the corrected files are committed on the results branch with this disclosure note. The preregistered procedure, criteria, and verdict rules are unchanged.

QPU Execution Record

Field	Value
Backend	ibm_fez (156-qubit Heron r2)
Job ID	d9dgp6kjeosc73fhigsg
Shots	4,000
Unique outcomes	256 (all $2^8 = 256$ bitstrings observed)
Submission timestamp (UTC)	2026-07-18T05:17:54.918200+00:00
Qubits	8, Hadamard on all, measure all

ProofRecord

```
{
  "schema": "witness.proofrecord/v1",
  "nonce": "2a06cecc238ddcf1e8c2774df2074d9bd3d3fdbcb7b1a3e6436281648aadad0ec",
  "job_id": "d9dgp6kjeosc73fhigsg",
  "backend": "ibm_fez",
  "calibration_hash": "b46857638b0c70526263589cf7d372a62c59ad9df8eac95c54cc68bf097bab0e",
  "timestamp_utc": "2026-07-18T05:17:54.918200+00:00",
  "raw_counts_hash": "09ec92f3c38a0ec59f30d95cefc7409736d4ccf919686bbcb4c7cfa4cad165e71",
  "context_id": "witness-1-primary"
}
```

Case Results

W1-C1 — Honest Verification

✓ PASS

Procedure: Recomputed nonce from stored raw counts + job ID + calibration snapshot; verified against ProofRecord nonce. Independently confirmed job ID and counts against the IBM Quantum provider API.

Check	Result
nonce_match	True
counts_hash_match	True
cal_hash_match	True
jobid_match	True
provider job_found	True
provider counts_match	True

Verification succeeded end-to-end. Meets PASS criterion.

W1-C2 — Substitution / Tamper Detection

✓ PASS

Procedure: Three independent sub-trials, one field substituted each time.

Sub-trial	Substitution	Forgery detected	Sub-verdict
(a)	raw_counts altered	True (nonce_match=False)	PASS
(b)	job_id altered	True (nonce_match=False)	PASS
(c)	calibration_snapshot altered	True (nonce_match=False)	PASS

All three substitutions detected as forged. Meets PASS criterion.

W1-C3 — Cross-Context Replay

✓ PASS

Procedure: Presented valid ProofRecord (context_id="witness-1-primary") to a second authorization context (context_id="witness-1-replay-context") using the ARK-457 replay-protection logic (5-dimensional context tuple, exact match required).

Check	Result
Original context decision	ALLOW
Replay context decision	DENY
Mismatched dimension(s)	session

Replay denied. Meets PASS criterion (DENY = PASS per preregistration).

Overall Verdict

✓ PASS

All three cases met their preregistered criteria. No deviations from the preregistered procedure, criteria, or thresholds.

What This Establishes (and Does Not)

Established within the tested construction:

- A ProofRecord nonce derived from QPU output (raw counts + job ID + calibration snapshot) can be recomputed and verified end-to-end against the provider's job record.

- Substitution of any single input field (counts, job ID, calibration) is detected as forgery under this construction.
- The ARK-457 replay-protection logic correctly denies a ProofRecord presented in a different authorization context.

Not established:

- Quantum randomness superiority over CSPRNG (no Bell test; device bias present).
- Universal security. Results are specific to this circuit, backend (ibm_fez), 8 qubits, 4,000 shots, calibration snapshot at execution time, and software harness.
- Production certification or cryptographic security guarantee.

Raw Artifact Hashes

Artifact	SHA-256
results/raw/raw_counts.json	09ec92f3c38a0ec59f30d95cefc7409736d4ccf919686bbc4c7cfa4cad165e71
results/raw/calibration.json	b46857638b0c70526263589cf7d372a62c59ad9df8eac95c54cc68bf097bab0e
results/proofrecord.json	see nonce field above

Related Works

- ARK Corpus (ExecutionProof series): <https://doi.org/10.5281/zenodo.21398675> (19 records / 17 IDs / 16 PASS / 2 FAIL / 1 gate-stop)
- ARK-457 (replay logic used in W1-C3): <https://doi.org/10.5281/zenodo.21421742>

Remnant Fieldworks RF Process. Preregistration-first. Results published as-is.

Zenodo published record: <https://zenodo.org/records/21424324>

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